

1 Traits predict forest phenological responses to photoperiod
2 more than temperature

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31 **Summary**

32 Shifts in the timing of spring phenology, like leafout, can have major ecosystem consequences. Under-
33 standing how species phenology fits within broader frameworks of plant strategies could help predict its

34 impacts on plant communities. Theory suggests that species with early spring phenology have a more
35 acquisitive growth strategy while later species are more conservative. Yet testing these predictions
36 has been slow given the high natural variability of leafout across space and time. Using controlled
37 environment studies, we reduce phenological variation into its component responses to winter and
38 spring conditions, and photoperiod, and test for trait relationships across 47 species sampled from
39 eight forests across North America. We find phenology connects to four plant traits via photoperiod
40 responses, but not temperature. Our results suggest photoperiod responses may be a critical compo-
41 nent of how spring phenology fits within growth strategies, which could help predict shifts in forest
42 growth with future climate change.

43 Keywords

44 Phenology, budburst, leafout, traits, forest communities, woody plants.

46 Introduction

47 Climate change is causing species phenologies—the timing of life history events—to shift, with widespread
48 advances being observed across the tree of life (Parmesan and Yohe, 2003; Hoegh-Guldberg et al., 2018).
49 This common phenological fingerprint, however, averages over high variability across species (Thack-
50 eray et al., 2016; Cohen et al., 2018; Kharouba et al., 2018), posing a challenge to accurate forecasts.

51
52 In plants, species variation can be explained, in part, by differences in growth strategies, which are
53 generally inferred from traits (Violle et al., 2007). Decades of research on plant traits have worked
54 to build predictive models of species responses to their environment (Green et al., 2022), which could
55 explain species-level variability in phenological responses. Phenology, however, has generally be ex-
56 cluded from plant trait research due to its high variability, even for the same species when observed
57 over different years or sites (Primack et al., 2009; Chuine et al., 2010). Within-species variation also
58 occurs across other plant traits (e.g. leaf and wood structure traits), but generally to a much smaller
59 scale compared to phenology (Violle et al., 2007). Yet increasing evidence that spring phenology is
60 much less variable when carefully measured (Zeng and Wolkovich, 2024; Ettinger et al., 2020) holds
61 promise to leverage existing frameworks to explain phenological variation and predict future changes.

62
63 Correlations between plant traits, growth strategies, and responses to environments have been synthe-
64 sized into several global frameworks, including the leaf economic spectrum (Wright et al., 2004) and
65 wood economic spectrum (Chave et al., 2009). These frameworks have identified key traits that ex-
66 hibit distinct gradients, ranging from acquisitive strategies—fast growing plants that produce cheaper
67 tissue—to conservative strategies—with plants that invest in long-lived tissue but slower growth rates
68 (Wright et al., 2004; Díaz et al., 2016). In temperate systems, changes in temperature in spring can
69 produce gradients in abiotic stress, including frost risk, soil nutrients, and light availability (Sakai and
70 Larcher, 1987; Gotelli and Graves, 1996; Augspurger, 2009), in addition to differences in biotic inter-

actions from herbivory or competition later in the season (Lopez et al., 2008; Wolkovich and Ettinger, 2014). Species that vary in their timing of leafout should therefore exhibit traits and growth strategies that allow them to tolerate or avoid these abiotic and biotic factors, but testing how phenology fits within major functional trait frameworks requires working across within- and between-species variation.

Phenology is generally observed in natural conditions where variation in the environment across space and time generates high variation in leafout, flowering and other phenological events (Chaine, 2000; Körner and Basler, 2010), which has stymied previous efforts to understand the role of phenology in existing trait frameworks. Increasing evidence, however, shows that phenology—especially spring phenology—is often far less variable when measured in experiments (Basler and Körner, 2014; Vitasse et al., 2009). Experiments in controlled environments (growth chambers) in particular can consistently decompose spring phenology for woody plants into its individual environmental cues (Chaine and Cour, 1999; Harrington and Gould, 2015; Flynn and Wolkovich, 2018): chilling (cool winter temperatures), forcing (warm spring temperatures), and photoperiod. Decades of experiments now show how these cues combine to determine early versus late spring phenology across species. While all species show strong responses to temperature cues, with chilling generally the strongest cue, later species tend to have larger responses to forcing, chilling and photoperiod (Ettinger et al., 2020). Within species, some evidence further suggests spatial gradients in climate may drive variation in responses to cues (Vitasse et al., 2018; Wang et al., 2025).

Here, we tested whether phenological variation was aligned with existing trait frameworks using data on spring budburst paired with a suite of traits that capture acquisitive to conservative growth strategies. We decomposed the high phenological variation in budburst date by using experiments to estimate chilling, forcing, and photoperiod cues for each species. We predicted that shrub and early spring species, which generally budburst before canopy closure (Augspurger, 2008; Vitasse et al., 2013), will have smaller responses to temperature and photoperiod and have traits associated with acquisitive growth. They would thus be shorter, with smaller trunks or stem diameters, and a lower investment in wood structure and leaf tissue, resulting in low branch wood density, diffuse-porous wood anatomy, and low leaf mass per area, but high leaf nitrogen content for a greater photosynthetic potential (Fig. 1). In contrast, we predicted species with later budburst (requiring more chilling, forcing, and longer photoperiods) to predominately include tree species that have more conservative growth strategies. These species should incur greater investments in long-lived tissue, with ring-porous wood anatomy, taller heights and greater diameter, denser wood and high leaf mass per area, but low leaf nitrogen content (Fig. 1). We further expected traits to vary spatially, with individuals at more northern sites being relatively smaller, with lower leaf mass per area relative to our southern sites, but follow similar responses in both our eastern and western transect. We used a joint-modeling approach to estimate the relationships between these plant traits and phenological responses to cues, while partitioning the variance from species- and population-level differences.

Materials and Methods

Field sampling

We combined *in situ* trait data with budburst data from two growth chamber cutting experiments conducted across eastern and western temperate deciduous forests in North America. We collected both suites of data from populations that span a latitudinal gradient of 4-6° for the eastern and western communities respectively. We took trait measurements from across eight populations, of which there were four eastern populations—Harvard Forest, Massachusetts, USA (42.55°N, 72.20°W), White Mountains, New Hampshire, USA (44.11°N, 52.14°W), Second College Grant, New Hampshire, USA (44.79°N, 50.66°W), and St. Hippolyte, Quebec, Canada (45.98°N, 74.01°W), and four western populations—E.C. Manning Park (49.06°N, 120.78°W), Sun Peaks (50.88°N, 119.89°W), Alex Fraser Research Forest (52.14°N, 122.14°W), and Smithers (54.78°N, 127.17°W), British Columbia (BC), Canada (Fig. 1). For the two growth chamber studies on budburst phenology, we collected cuttings from the most southern and northern populations in both our western and eastern transect ($n_{pop}=4$).

Across our eight populations, we measured 47 unique species of which 28 were in our eastern transect and 22 in our western transect (Table S1). At each site, we sampled the dominant deciduous species in both the understory and canopy layer. At our eastern community, we sampled 13 shrub species and 15 species of tree, while at our western sites we sampled 18 shrub and 4 tree species, of which three tree species occurred in both our eastern and western transect.

Functional traits

We measured all traits in the summer prior to each growth chamber study (eastern transect: 8-25 June 2015, western transect: 29 May to 30 July 2019), following full leafout but before budset. At each population and for each species, we measured a total of five traits from 1-10 healthy adult individuals: height, diameter of the main trunk or stem (hereafter referred to as diameter), wood density, leaf mass per area, and the percent leaf nitrogen content. We also obtained xylem structure data from the WSL xylem database (Schweingruber and Landolt, 2010) for 72.3% of our species.

We measured traits in accordance to the methods discussed by Pérez-Harguindeguy et al. (2013). We calculated tree height using trigonometric methods and used a base height of 1.37 m to measure tree diameter (Magarik et al., 2020). For shrub heights, we measured the distance from the ground to the height of the top foliage and measured stem diameters at approximately 1 cm above ground-level. All stem and leaf samples were kept cool during transport and measurements of leaf area and stem volume taken within 3 and 12 hours of sample collection respectively. To measure wood density, we collected a 10 cm sample of branch wood, taken close to the base of the branch at the stem and calculated stem volume using the water displacement method. For our leaf traits, we haphazardly selected and sampled five, fully expanded, and hardened leaves, with no to minimal herbivore damage. We took a high resolution scan of each leaf using a flatbed scanner and estimated leaf area using the ImageJ software (version 2.0.0).

148 Growth chamber study

149 For our growth chamber studies, we collected dormant branch cuttings from our highest and lowest
 150 latitude populations in each transect (total of four sites), with sampling in our eastern study occurring
 151 from 20-28 January 2015 and sampling for our western study from 19-28 October 2019. Dormant
 152 branch cuttings have been repeatedly shown to approximate whole plant responses in budburst (Vi-
 153 tasse and Basler, 2014), allowing us to estimate responses to environmental cues. In both our eastern
 154 and western controlled environment study, we included two temperature treatments and photoperiod,
 155 for a total of eight distinct treatments (Fig. 1). Our treatments included two levels of chilling—with
 156 our eastern study having no additional chilling or 30 days at 4°C, and our western study 21 days or
 157 56 days of chilling at 4°C, with all non-field chilling occurring under dark conditions. Our forcing
 158 treatments included either a cool regime of 15:5°C or a warm regime of 20:10°C, and our photoperiod
 159 treatments consisted of either a 8 or 12 hour photoperiod. These treatment levels, while greater than
 160 the conditions individuals are likely to experience in the field, are comparable to treatments commonly
 161 used in controlled environment experiments and allow us to make similar inferences as studies that
 162 apply site-specific conditions (Ettinger et al., 2020, 2021). We recorded budburst stages of each sample
 163 every 1-3 days for up to four months, defining the day of budburst as the day of budbreak or shoot
 164 elongation (denoted as code 07 by Finn et al., 2007). For a more detailed discussion of study sample
 165 collection and methods see Flynn and Wolkovich (2018) for details on our eastern study and Loughnan
 166 and Wolkovich (in review) for details on our western study.

168 Statistical Analysis

169 We combined our *in situ* trait data with budburst data from the controlled environment experiments
 170 through a joint Bayesian model for each trait. This approach improves upon previous analyses of trait
 171 correlations, as it allows us to carry through uncertainty between trait and phenology estimates—and
 172 better partitions the drivers of variation in species phenologies.

173
 174 Our joint model consists of two parts. The first is a hierarchical linear model, which partitions the vari-
 175 ation of individual observations (i) of a given trait value (Y_{trait}) to account for the effects of species
 176 (j), population-level differences arising from transects, latitude, as well as the interaction between
 177 transects and latitude ($\text{transect} \cdot \text{latitude}$), and finally, residual variation or ‘measurement error’ (σ_m^2).

$$Y_{\text{trait}_{i,j}} \sim \text{Normal}(\mu_{i,j}, \sigma_m^2) \quad (1)$$

$$\mu_{i,j} = \alpha_{\text{grand trait}} + \alpha_{\text{trait}_j} + \beta_{\text{transect}} \times \text{transect} + \beta_{\text{latitude}} \times \text{latitude} + \beta_{\text{transect} \cdot \text{latitude}} \times (\text{transect} \cdot \text{latitude}) \quad (2)$$

179 Where α_{trait} is an element of the normal random vector:

$$\alpha_{\text{trait}} \begin{bmatrix} \alpha_{\text{trait}_1} \\ \alpha_{\text{trait}_2} \\ \dots \\ \alpha_{\text{trait}_n} \end{bmatrix} \text{ such that } \alpha_{\text{trait}} \sim \text{Normal}(0, \sigma_{\text{trait}}^2)$$

180 We included transect as a dummy variable (0/1) and latitude as a continuous variable and modeled
 181 traits in their original units, with the exception of leaf mass per area which was rescaled by 100 and
 182 wood density which was rescaled by 10 for numeric stability. Our model also included partial pooling
 183 for species—which controls for uneven sampling and trait variability. These species-level estimates are
 184 then predictors for each cue ($\beta_{\text{chilling},j}$, $\beta_{\text{forcing},j}$, $\beta_{\text{photoperiod},j}$).

185

$$\begin{aligned} \beta_{\text{chilling}_j} &= \alpha_{\text{chilling},j} + \beta_{\text{trait.chilling}} \times \alpha_{\text{trait}_j} \\ \beta_{\text{forcing}_j} &= \alpha_{\text{forcing},j} + \beta_{\text{trait.forcing}} \times \alpha_{\text{trait}_j} \\ \beta_{\text{photoperiod}_j} &= \alpha_{\text{photoperiod},j} + \beta_{\text{trait.photoperiod}} \times \alpha_{\text{trait}_j} \end{aligned} \quad (3)$$

186 In addition to the species-level estimates, this part of our model estimates the overall effect of each
 187 trait on each cue ($\beta_{\text{trait.chilling}}$, $\beta_{\text{trait.forcing}}$, $\beta_{\text{trait.photoperiod}}$). From this we could estimate how well
 188 traits explain species-level differences—by estimating the the species-level cue variation not explained
 189 by traits ($\alpha_{\text{chilling},j}$, $\alpha_{\text{forcing},j}$, $\alpha_{\text{photoperiod},j}$) and individual species responses to cues (*chilling*, *forcing*,
 190 *photoperiod*, respectively). Finally, our model estimates the residual budburst variation across species
 191 (α_{pheno_j}), observations (σ_d^2), and the variation in cues not attributed to the trait using partial pooling,
 192 with the day of observed budburst ($Y_{\text{pheno}_{i,j}}$) reflecting the additive effects of this variation and the
 193 differences in species response to the experimental treatments.

$$Y_{\text{pheno}_{i,j}} \sim \text{Normal}(\mu_{i,j}, \sigma_d^2) \quad (4)$$

194 with

$$\mu_{i,j} = \alpha_{\text{pheno}_j} + \beta_{\text{chilling}_j} \cdot \text{chilling} + \beta_{\text{forcing}_j} \cdot \text{forcing} + \beta_{\text{photoperiod}_j} \cdot \text{photoperiod} \quad (5)$$

195 where α_{pheno_j} , $\alpha_{\text{chilling}_j}$, $\alpha_{\text{forcing}_j}$, and $\alpha_{\text{photoperiod}_j}$ are elements of the normal random vectors:

$$\boldsymbol{\alpha}_{\text{pheno}} = \begin{bmatrix} \alpha_{\text{pheno}_1} \\ \alpha_{\text{pheno}_2} \\ \dots \\ \alpha_{\text{pheno}_n} \end{bmatrix} \text{ such that } \boldsymbol{\alpha}_{\text{pheno}} \sim \text{Normal}(\mu_{\text{pheno}}, \sigma_{\text{pheno}}^2) \quad (6)$$

$$\boldsymbol{\alpha}_{\text{chilling}} = \begin{bmatrix} \alpha_{\text{chilling}_1} \\ \alpha_{\text{chilling}_2} \\ \dots \\ \alpha_{\text{chilling}_n} \end{bmatrix} \text{ such that } \boldsymbol{\alpha}_{\text{chilling}} \sim \text{Normal}(\mu_{\text{chilling}}, \sigma_{\text{chilling}}^2) \quad (7)$$

$$\boldsymbol{\alpha}_{\text{forcing}} = \begin{bmatrix} \alpha_{\text{forcing}_1} \\ \alpha_{\text{forcing}_2} \\ \dots \\ \alpha_{\text{forcing}_n} \end{bmatrix} \text{ such that } \boldsymbol{\alpha}_{\text{forcing}} \sim \text{Normal}(\mu_{\text{forcing}}, \sigma_{\text{forcing}}^2) \quad (8)$$

$$\boldsymbol{\alpha}_{\text{photoperiod}} = \begin{bmatrix} \alpha_{\text{photoperiod}_1} \\ \alpha_{\text{photoperiod}_2} \\ \dots \\ \alpha_{\text{photoperiod}_n} \end{bmatrix} \text{ such that } \boldsymbol{\alpha}_{\text{photoperiod}} \sim \text{Normal}(\mu_{\text{photoperiod}}, \sigma_{\text{photoperiod}}^2) \quad (9)$$

196 We modeled each trait individually, with the exception of ring-porosity, which we compared across
 197 species using the posterior estimates of our wood density model, allowing us to account for inherent
 198 differences in wood anatomy across species and growth form. We included all three cues (chilling,
 199 forcing, and photoperiod) as continuous variables in our model. We converted chilling temperatures
 200 to total chill portions, including both the chilling experienced in the field prior to sampling and during
 201 the experiment. For this we used local weather station data and the chillR package (v. 0.73.1, Luedel-
 202 ing, 2020). To account for differences in thermoperiodicity between our eastern and western growth
 203 chamber studies (Buonaiuto et al., 2023), we also converted forcing temperatures to mean daily tem-
 204 peratures for each treatment. Finally, we z -scored each cue and site using two standard deviations to
 205 allow direct comparisons between results across parameters (Gelman, 2008).

206

207 We used trait-specific priors that were weakly informative. We validated our choice of priors using
 208 prior predictive checks and confirmed model stability under wider priors. All models were coded
 209 in the Stan programming language for Bayesian models using the rstan package (Stan Development
 210 Team, 2018) in R version 4.3.1 (R Development Core Team, 2017). All models met basic diagnostic
 211 checks, including no divergences, high effective sample sizes (n_{eff}) that exceeded 10% of the number of
 212 iterations, and $\hat{R}$ values close to 1. We report our model estimates as the mean values with 90% uncer-
 213 tainty intervals (UI), interpreting parameter estimates with intervals that include zero to have no effect.

214

Results

Across our eight populations, we measured traits of 1428 unique individuals between the two transects across our five *in situ* traits: height ($n = 1317$), diameter ($n = 1220$), wood density ($n = 1359$), leaf mass per area ($n = 1345$), leaf nitrogen content ($n = 1351$). Across our two growth chamber studies, we made observations of 4211 branch cuttings, with our observations of budburst spanning 82 and 113 days for our eastern and western studies respectfully.

We found that four of our five traits had a strong relationship with photoperiod, but not always in the direction we predicted. Taller species with larger trunk diameters and leaves with high nitrogen content did have larger responses with longer photoperiods (Fig. 3 c, i, o; Tables S3, S4, S7), as we predicted. But, contrary to our expectation, species with denser, high leaf mass per area leaves had smaller photoperiod responses, allowing them to potentially budburst earlier, under shorter photoperiods (Fig. 3f, Table S6).

Temperature cues ($\beta_{\text{trait.chilling}}$ and $\beta_{\text{trait.forcing}}$) exhibited no relationships with individual traits, but we found the individual importance of our three cues on budburst was not consistent after accounting for the effects of specific leaf or wood traits.

Chilling (β_{chilling}) had the strongest effect in our models of height (-13.4 days per standardized chill portions, UI: -17.2, -9.9), diameter (-12.5 days per standardized chill portions, UI: -16.2, -8.6), wood density (-20.9 days per standardized chill portions, UI: -33.2, -9.8), and leaf nitrogen content (-35.1 days per standardized chill portions, UI: -68.1, -4.1), with more chilling advancing budburst. Our model of leaf mass per area estimated photoperiod ($\beta_{\text{photoperiod}}$) as the strongest cue (-14.0 days per standardized photoperiod, UI: -23.1, -3.5). But contrary to previous controlled environment experiments, budburst timing showed strong responses to all three cues in only two of our trait models. In accounting for the effects of height and diameter (Tables S3, S4), budburst was estimated to respond to all three cues. But in our models accounting for the effects of wood density and leaf nitrogen content, temperature cues alone shaped budburst timing (Tables S5, S7), and in our model of leaf mass per area, budburst was estimated to respond to just photoperiod cues (Table S6).

Of our four trait models, we found clear gradients in species timing between trees and shrubs in our model height and diameter. Species height exhibited large correlations between budburst timing and trait values, with earlier estimates of budburst for shrubs (with a mean day of budburst of 10)—especially under greater cues—and later budburst estimates for trees (with a mean day of budburst of 17.3, Fig. S1). But this was not the case for the leaf traits. Leaf nitrogen content, in particular, showed no distinct separation between shrub and tree functional groups (Fig. S1). We also found no differences between xylem structure and their responses to cues (Fig. 4).

Most of our traits varied across latitude within each transect and had strong interactive effects between transect and latitude (Fig. 2). Leaf nitrogen content varied with latitude alone, with low latitude sites on both our eastern and western transects having more nitrogen rich leaves than sites at higher latitudes (-0.1 percent per degree latitude, UI: -0.2, 0.0, Table S7). Plants also had wider trunk or stem

diameters at higher latitudes in both eastern and western sites (0.5 cm per degree latitude, UI: 0.1, 0.9), with a larger effect in eastern sites (Fig. 2d). Relationships with height, woody density and leaf mass per area were, however, more complex. Plants heights and wood density increased with increasing latitude in western sites, but decreased with latitude in eastern sites (-0.2 m per degree latitude, UI: -0.4, 0.0 for our height model and -0.01 g/cm³ per degree latitude, UI: -0.02, 0.0 for our wood density model; Fig. 2 a and c). For leaf mass per area we found the opposite relationship (decreasing leaf mass per area with increasing latitudes in western sites, while it increased with latitude in our eastern sites (0.005 g/cm² per degree latitude, UI: 0.004, 0.006; Fig. 2d). In addition to the differences we found across sites, we also observed large differences between individual species, which varied up to 7 fold for some traits (Fig. 3).

Discussion

Using a joint modeling approach to understand how phenology may fit within acquisitive to conservative plant strategies, we found that photoperiod was the only cue related to our suite of leaf and wood traits. While budburst responses to photoperiod are often smaller than temperature responses (chilling and forcing; Laube et al., 2014; Zohner et al., 2016; Flynn and Wolkovich, 2018), our results suggest photoperiod may be the most important cue in linking spring phenology to functional traits. To accurately predict these responses thus requires considering additional traits and variation across space, as we observed differences in the relative importance of cues across our trait models and between our eastern and western transects with latitude. The observed spatial differences in trait variation may be due to differences in the community assemblages, as our western community was more shrub dominated, with shorter plants with less dense branch wood, suggesting a more acquisitive growth strategy to allow species to utilize resources early in the season before canopy closure.

Cues and functional traits

We found only partial support for our prediction that species with acquisitive traits—particularly small trees with low wood density, low leaf mass per area, and high leaf nitrogen content—would have smaller temperature and photoperiod responses (associated with early budburst). Species with smaller heights and diameters did have smaller photoperiod responses, but—contrary to our prediction—species with less dense leaves showed larger responses to photoperiod, while leaves with high nitrogen content had stronger photoperiod responses.

None of our focal traits showed a relationship with temperature (chilling or forcing), which may be due to selection on other physiological processes. Many of our traits are associated with one or more ecological function (Wright et al., 2004; Pérez-Harguindeguy et al., 2013; Reich, 2014). In particular, leaf mass per area is known to correlate with traits like leaf lifespan or decomposition rates in addition to light capture (De La Riva et al., 2016). In accounting for the relationships between other traits and budburst cues, we also found the relative importance of temperature and photoperiod on budburst timing varied, suggesting other traits may play a role in shaping phenologies.

Decades of previous phenology research have found budburst timing to be primarily driven by temperature (chilling and forcing) and weakly by photoperiod (Chuine et al., 2010; Basler and Körner, 2014; Laube et al., 2014). Yet we found no traits that correlated with responses to temperature, suggesting other drivers may impact leaf and structural traits in temperate forests. One potential abiotic driver we did not consider is soil moisture, which covaries with a number of traits, including leaf mass per area, as higher leaf area allows plants to reduce evaporation under dry conditions (De La Riva et al., 2016). Soil moisture also shapes other phenological events in woody plants, including radial growth phenology and shoot elongation (Cabon et al., 2020; Peters et al., 2021). Though temperate forests are generally moist compared to other systems, previous studies suggests soil moisture can shape spring phenology across xeric systems as well (Crimmins et al., 2011; Park, 2014; Ettinger et al., 2019), thus including soil moisture in future studies could help understand how species growth strategies correlate with phenology.

307

Our finding that budburst was not related to wood structure or wood density contrasts the findings of previous work linking these traits. Previous studies have found some evidence that trees with diffuse-porous wood structure leafout earlier than species with ring-porous structures (Lechowicz, 1984; Panchen et al., 2014; Yin et al., 2016; Osada, 2017; Savage et al., 2022). Using wood density as an alternative measure of wood structure (wood density positively correlates with xylem resistance to embolism, Hacke et al., 2001), we did not find an association between our three phenological cues and xylem structure, despite our data including similar temperate forest species to previous studies. This could be because we captured additional variation by sampling many individuals for each species across a large spatial gradient. In particular, we found larger wood densities at higher latitudes in our western transect, which could be caused by the differences in winter conditions experienced across this transect. Higher wood densities are especially favorable in communities that experience greater horizontal stress from wind and downward pressure from snow (MacFarlane and Kane, 2017; MacFarlane, 2020). Relationships with wood density may also be more apparent over a larger span of values, as our observations varied from 0.2 to 0.6 g/cm³, which is smaller than seen in studies that span a more global distribution (Galvão et al., 2021; Savage et al., 2022; Mo et al., 2024).

323

In addition to our study providing insight into how trait-budburst relationships vary with latitude, our sites also spanned North America, encompassing a gradient of 55° in longitude. At this continental scale we found correlations between phenology and traits for woody species in temperate forests, but these may shift with data that expands across biomes and more plant functional groups. Indeed, both the leaf and wood economic spectra find that large variation in traits across species arises from large global datasets and include both temperate and tropical species (Wright et al., 2004; Díaz et al., 2016; Chave et al., 2009) and functional groups that span from trees to grasses (Wright et al., 2004). Global patterns in trait variation may also be absent at smaller spatial scales (Wright and Sutton-Grier, 2012; Messier et al., 2017*a,b*), suggesting that comparisons of trait-phenology relationships at larger spatial scales or across more diverse pools of species may better align with the patterns predicted by existing economic spectra.

334

335

336 In comparing spring phenology of mainly temperate woody species, our results support previous work
 337 through which a global meta-analysis found that spring phenology may correlate with height and leaf
 338 mass per area (Loughnan et al., 2025). At both the global and continental scales, trees with taller
 339 heights leafed out with longer photoperiods. We also found species with high specific leaf area—which
 340 is the inverse of leaf mass per area and thus equivalent to low values—exhibited large responses to
 341 photoperiod (Loughnan et al., 2025). The consistency of these results, despite the differences in the two
 342 spatial scales of these datasets, provides further support that woody species responses to photoperiod
 343 may be part of a larger plant strategy.

344

345 **Functional traits predict climate change responses**

346 Our results offer novel insights into how broader correlations between growth strategies and phenologi-
 347 cal cues can help predict responses in woody plant communities with climate change. As temperatures
 348 rise, particularly at higher latitudes (Hoegh-Guldberg et al., 2018), warmer winter and spring temper-
 349 atures may select for earlier budburst in some species. But, since photoperiod will remain fixed, our
 350 observed relationships between photoperiod and other traits has the potential to limit species abilities
 351 to track temperatures. This could constrain the extent to which some species growth will advance with
 352 climate change. Our results suggest that these effects will likely be greater for taller species or canopy
 353 trees and species with relatively low leaf mass per area. These constraints could have cascading effects
 354 on forest communities, as variable species responses to increasing temperatures further alter species
 355 growth strategies and their interactions with competitors or herbivores within their communities.

356

357 Our findings of correlations between phenology and other commonly measured traits highlight how
 358 accurate forecasts of future changes in phenology could benefit from accounting for the response of
 359 other traits to climate change. Across temperature and precipitation gradients, leaf size and shape also
 360 change, as species shift to conserve water and mitigate the effects of transpiration under higher tem-
 361 peratures (De La Riva et al., 2016). These changes could impact species photosynthetic potential and
 362 ultimately ecosystem services, such as carbon sequestration. While phenological research has focused
 363 on forecasting responses to temperature, the correlation of other traits with photoperiod suggests it
 364 is also an important cue. Considering the tradeoffs and differences in cues that together shape plants
 365 growth strategies may yield more accurate forecasts of species phenology and community dynamics
 366 under future climates.

367

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376

377 Competing interests

378 The authors have no competing interests.
379

380 Author contributions

381 Both D.L. and E.M.W. conceived the study, D.L., F.A.J. and E.M.W. all contributed to the analysis
382 and code and contributed to the writing and revision of the manuscript. DL led field and experimental
383 data collection.
384

385 Data availability

386 The phenology data used in this study is published on the Knowledge Network of Biocomplexity
387 for both the eastern transect (urn:uuid:efeccd2b-b850-4b48-b721-9bed8040562e) and western transect
388 (doi:10.5063/F1W66J82). Trait data from the eastern transect is available through the Harvard Forest
389 Data Archive (doi:10.6073/pasta/dc68c63764f0c5edf6f09e7a60a0468a). Trait data from the western
390 transect will be published on the Knowledge Network of Biocomplexity.

391 References

- 392 Augspurger, C. K. 2008. Early spring leaf out enhances growth and survival of saplings in a temperate
393 deciduous forest. *Oecologia* 156:281–286.
- 394 ———. 2009. Spring 2007 warmth and frost: phenology, damage and refoliation in a temperate
395 deciduous forest. *Functional Ecology* 23:1031–1039.
- 396 Basler, D., and C. Körner. 2014. Photoperiod and temperature responses of bud swelling and bud
397 burst in four temperate forest tree species. *Tree Physiology* 34:377–388.
- 398 Buonaiuto, D. M., E. M. Wolkovich, and M. J. Donahue. 2023. Experimental designs for testing
399 the interactive effects of temperature and light in ecology: The problem of periodicity. *Functional*
400 *Ecology* 37:1747–1756.
- 401 Cabon, A., L. Fernández-de-Uña, G. Gea-Izquierdo, F. C. Meinzer, D. R. Woodruff, J. Martínez-
402 Vilalta, and M. De Cáceres. 2020. Water potential control of turgor-driven tracheid enlargement in
403 Scots pine at its xeric distribution edge. *New Phytologist* 225:209–221.

Chave, J., D. Coomes, S. Jansen, S. L. Lewis, N. G. Swenson, and A. E. Zanne. 2009. Towards a worldwide wood economics spectrum. *Ecology Letters* 12:351–366.

Chuine, I. 2000. A unified model for budburst of trees. *Journal of Theoretical Biology* 207:337–347.

Chuine, I., and P. Cour. 1999. Climatic determinants of budburst seasonality in four temperate-zone tree species. *New Phytologist* 143:339–349.

Chuine, I., X. Morin, and H. Bugmann. 2010. Warming, photoperiods, and tree phenology. *Science* 329:277–278.

Cohen, J. M., M. J. Lajeunesse, and J. R. Rohr. 2018. A global synthesis of animal phenological responses to climate change. *Nature Climate Change* 8:224–228.

Crimmins, T. M., M. A. Crimmins, and C. D. Bertelsen. 2011. Onset of summer flowering in a ‘Sky Island’ is driven by monsoon moisture. *New Phytologist* 191:468–479.

De La Riva, E. G., M. Olmo, H. Poorter, J. L. Uberta, and R. Villar. 2016. Leaf Mass per Area (LMA) and Its Relationship with Leaf Structure and Anatomy in 34 Mediterranean Woody Species along a Water Availability Gradient. *PLOS ONE* 11:e0148788.

Díaz, S., J. Kattge, J. H. Cornelissen, I. J. Wright, S. Lavorel, S. Dray, B. Reu, M. Kleyer, C. Wirth, I. Colin Prentice, E. Garnier, G. Bönsch, M. Westoby, H. Poorter, P. B. Reich, A. T. Moles, J. Dickie, A. N. Gillison, A. E. Zanne, J. Chave, S. Joseph Wright, S. N. Sheremet Ev, H. Jactel, C. Baraloto, B. Cerabolini, S. Pierce, B. Shipley, D. Kirkup, F. Casanoves, J. S. Joswig, A. Günther, V. Falczuk, N. Rüger, M. D. Mahecha, and L. D. Gorné. 2016. The global spectrum of plant form and function. *Nature* 529:167–171.

Ettinger, A. K., D. M. Buonaiuto, C. J. Chamberlain, I. Morales-Castilla, and E. M. Wolkovich. 2021. Spatial and temporal shifts in photoperiod with climate change. *New Phytologist* 230:462–474.

Ettinger, A. K., C. J. Chamberlain, I. Morales-Castilla, D. M. Buonaiuto, D. F. B. Flynn, T. Savas, J. A. Samaha, and E. M. Wolkovich. 2020. Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change* 10:1137–1142.

Ettinger, A. K., I. Chuine, B. I. Cook, J. S. Dukes, A. M. Ellison, M. R. Johnston, A. M. Panetta, C. R. Rollinson, Y. Vitasse, and E. M. Wolkovich. 2019. How do climate change experiments alter plot-scale climate? *Ecology Letters* 22:748–763.

Finn, G. A., A. E. Straszewski, and V. Peterson. 2007. A general growth stage key for describing trees and woody plants. *Annals of Applied Biology* 151:127–131.

Flynn, D. F. B., and E. M. Wolkovich. 2018. Temperature and photoperiod drive spring phenology across all species in a temperate forest community. *New Phytologist* 219:1353–1362.

Galvão, F. G., A. L. Alves De Lima, C. Candeia De Oliveira, V. F. Da Silva, and M. J. N. Rodal. 2021. The importance of wood density in determining the phenology of tree species in a coastal rain forest. *Biotropica* 53:1134–1141.

439 Gelman, A. 2008. Scaling regression inputs by dividing by two standard deviations. *Statistics in*
440 *Medicine* 27:2865–2873.

441 Gotelli, N. J., and G. R. Graves. 1996. The temporal niche. Pages 95–112 *in* *Null Models In Ecology*.
442 Smithsonian Institution Press, Washington, D. C.

443 Green, S. J., C. B. Brookson, N. A. Hardy, and L. B. Crowder. 2022. Trait-based approaches to
444 global change ecology: moving from description to prediction. *Proceedings of the Royal Society B:*
445 *Biological Sciences* 289:1–10.

446 Hacke, U. G., J. S. Sperry, W. T. Pockman, S. D. Davis, and K. A. McCulloh. 2001. Trends in wood
447 density and structure are linked to prevention of xylem implosion by negative pressure. *Oecologia*
448 126:457–461.

449 Harrington, C. A., and P. J. Gould. 2015. Tradeoffs between chilling and forcing in satisfying dormancy
450 requirements for Pacific Northwest tree species. *Frontiers in Plant Science* 6:1–12.

451 Hoegh-Guldberg, O., D. Jacob, M. Taylor, M. Bindi, S. Brown, I. Camilloni, A. Diedhiou, R. Djalante,
452 K. Ebi, F. Engelbrecht, J. Guiot, Y. Hijioka, S. Mehrotra, A. Payne, S. Seneviratne, A. Thomas,
453 R. Warren, and G. Zhou. 2018. Impacts of 1.5 °C Global Warming on Natural and Human Systems.
454 In: *Global Warming of 1.5 °C. An IPCC Special Report on the impacts of global warming of 1.5 °C*
455 *above pre-industrial levels and related global greenhouse gas emission pathways, in the context of .*
456 *Tech. rep.*, Cambridge University Press, Cambridge, UK and New York, NY, USA.

457 Kharouba, H. M., J. Ehrlén, A. Gelman, K. Bolmgren, J. M. Allen, S. E. Travers, and E. M. Wolkovich.
458 2018. Global shifts in the phenological synchrony of species interactions over recent decades. *Pro-*
459 *ceedings of the National Academy of Sciences* 115:5211–5216.

460 Körner, C., and D. Basler. 2010. Phenology Under Global Warming. *Science* 327:1461–1463.

461 Laube, J., T. H. Sparks, N. Estrella, J. Höfler, D. P. Ankerst, and A. Menzel. 2014. Chilling outweighs
462 photoperiod in preventing precocious spring development. *Global Change Biology* 20:170–182.

463 Lechowicz, M. J. 1984. Why Do Temperate Deciduous Trees Leaf Out at Different Times? *Adaptation*
464 *and Ecology of Forest Communities*. *The American Naturalist* 124:821–842.

465 Lopez, O. R., K. Farris-Lopez, R. A. Montgomery, and T. J. Givnish. 2008. Leaf phenology in relation
466 to canopy closure in southern Appalachian trees. *American Journal of Botany* 95:1395–1407.

467 Loughnan, D., F. A. Jones, G. Legault, C. J. Chamberlain, D. M. Buonaiuto, A. K. Ettinger, M. Gar-
468 ner, D. S. Sodhi, and E. M. Wolkovich. 2025. Budburst timing within a functional trait framework.
469 *Journal of Ecology* 113:3787–3798.

470 Loughnan, D., and E. M. Wolkovich. in review. Temporal assembly of woody plant communities shaped
471 equally by evolutionary history as by current environments .

472 Luedeling, E. 2020. *chillR: Statistical Methods for Phenology Analysis in Temperate Fruit Trees*.
473 <https://CRAN.R-project.org/package=chillR>.

474 MacFarlane, D. W. 2020. Functional Relationships Between Branch and Stem Wood Density for
475 Temperate Tree Species in North America. *Frontiers in Forests and Global Change* 3:1–16.

476 MacFarlane, D. W., and B. Kane. 2017. Neighbour effects on tree architecture: functional trade-offs
477 balancing crown competitiveness with wind resistance. *Functional Ecology* 31:1624–1636.

478 Magarik, Y. A., L. A. Roman, and J. G. Henning. 2020. How should we measure the dbh of multi-
479 stemmed urban trees? *Urban Forestry & Urban Greening* 47:1–11.

480 Messier, J., M. J. Lechowicz, B. J. McGill, C. Violle, and B. J. Enquist. 2017*a*. Interspecific integration
481 of trait dimensions at local scales: the plant phenotype as an integrated network. *Journal of Ecology*
482 105:1775–1790.

483 Messier, J., B. J. McGill, B. J. Enquist, and M. J. Lechowicz. 2017*b*. Trait variation and integration
484 across scales: is the leaf economic spectrum present at local scales? *Ecography* 40:685–697.

485 Mo, L., T. W. Crowther, D. S. Maynard, and e. a. Van Den Hoogen. 2024. The global distribution
486 and drivers of wood density and their impact on forest carbon stocks. *Nature Ecology & Evolution*
487 8:2195–2212.

488 Osada, N. 2017. Relationships between the timing of budburst, plant traits, and distribution of 24
489 coexisting woody species in a warm-temperate forest in Japan. *American Journal of Botany* 104:550–
490 558.

491 Panchen, Z. A., R. B. Primack, B. Nordt, E. R. Ellwood, A. Stevens, S. S. Renner, C. G. Willis,
492 R. Fahey, A. Whittmore, Y. Du, and C. C. Davis. 2014. Leaf out times of temperate woody
493 plants are related to phylogeny, deciduousness, growth habit and wood anatomy. *New Phytologist*
494 203:1208–1219.

495 Park, I. W. 2014. Impacts of differing community composition on flowering phenology throughout warm
496 temperate, cool temperate and xeric environments. *Global Ecology and Biogeography* 23:789–801.

497 Parmesan, C., and G. Yohe. 2003. A globally coherent fingerprint of climate change impacts across
498 natural systems. *Nature* 421:37–42.

499 Pérez-Harguindeguy, N., S. Díaz, E. Garnier, S. Lavorel, H. Poorter, P. Jaureguiberry, M. S. Bret-
500 Harte, W. K. Cornwell, J. M. Craine, D. E. Gurvich, C. Urcelay, E. J. Veneklaas, P. B. Reich,
501 L. Poorter, I. J. Wright, P. Ray, L. Enrico, J. G. Pausas, A. C. de Vos, N. Buchmann, G. Funes,
502 F. Quétier, J. G. Hodgson, K. Thompson, H. D. Morgan, H. ter Steege, M. G. A. van der Heijden,
503 L. Sack, B. Blonder, P. Poschlod, M. V. Vaieretti, G. Conti, A. C. Staver, S. Aquino, and J. H. C.
504 Cornelissen. 2013. New handbook for standardized measurement of plant functional traits worldwide.
505 *Australian Journal of Botany* 61:167–234.

506 Peters, R. L., K. Steppe, H. E. Cuny, D. J. De Pauw, D. C. Frank, M. Schaub, C. B. Rathgeber,
507 A. Cabon, and P. Fonti. 2021. Turgor – a limiting factor for radial growth in mature conifers along
508 an elevational gradient. *New Phytologist* 229:213–229.

509 Primack, R. B., I. Ibáñez, H. Higuchi, S. D. Lee, A. J. Miller-Rushing, A. M. Wilson, and J. A. Silander. 2009. Spatial and interspecific variability in phenological responses to warming temperatures. 510 Biological Conservation 142:2569–2577.

511

512 R Development Core Team. 2017. R: A language and environment for statistical computing.

513 Reich, P. B. 2014. The world-wide ‘fast–slow’ plant economics spectrum: a traits manifesto. Journal 514 of Ecology 102:275–301.

515 Sakai, A., and W. Larcher. 1987. Frost Survival of Plants: Responses and adaptation to freezing stress. 516 Springer-Verlag, Berlin, Heidelberg.

517 Savage, J. A., T. Kiecker, N. McMann, D. Park, M. Rothendler, and K. Mosher. 2022. Leaf out time 518 correlates with wood anatomy across large geographic scales and within local communities. New 519 Phytologist 235:953–964.

520 Schweingruber, F., and W. Landolt. 2010. The xylem database.

521 Stan Development Team. 2018. RStan: the R interface to Stan. R package version 2.17.3.

522 Thackeray, S. J., P. A. Henrys, D. Hemming, J. R. Bell, M. S. Botham, S. Burthe, P. Helaouet, 523 D. G. Johns, I. D. Jones, D. I. Leech, E. B. MacKay, D. Massimino, S. Atkinson, P. J. Bacon, 524 T. M. Brereton, L. Carvalho, T. H. Clutton-Brock, C. Duck, M. Edwards, J. M. Elliott, S. J. Hall, 525 R. Harrington, J. W. Pearce-Higgins, T. T. Høye, L. E. Kruuk, J. M. Pemberton, T. H. Sparks, 526 P. M. Thompson, I. White, I. J. Winfield, and S. Wanless. 2016. Phenological sensitivity to climate 527 across taxa and trophic levels. Nature 535:241–245.

528 Violle, C., M. Navas, D. Vile, E. Kazakou, C. Fortunel, I. Hummel, and E. Garnier. 2007. Let the 529 concept of trait be functional! Oikos 116:882–892.

530 Vitasse, Y., and D. Basler. 2014. Is the use of cuttings a good proxy to explore phenological responses 531 of temperate forests in warming and photoperiod experiments? Tree Physiology 34:174–183.

532 Vitasse, Y., G. Hoch, C. F. Randin, A. Lenz, C. Kollas, J. F. Scheepens, and C. Körner. 2013. 533 Elevational adaptation and plasticity in seedling phenology of temperate deciduous tree species. 534 Oecologia 171:663–678.

535 Vitasse, Y., A. Josée, A. Kremer, R. Michalet, and S. Delzon. 2009. Responses of canopy duration to 536 temperature changes in four temperate tree species : relative contributions of spring and autumn 537 leaf phenology. Oecologia 161:187–198.

538 Vitasse, Y., C. Signarbieux, and Y. H. Fu. 2018. Global warming leads to more uniform spring 539 phenology across elevations. PNAS 115:1004–1008.

540 Wang, N., Z. Wu, Y. Gong, Y. Cui, and Y. H. Fu. 2025. Background climate determines the response 541 of spring leaf-out to climate change—Results from a national-scale twig-cutting experiment. Journal 542 of Ecology 113:2981–2991.

- 543 Wolkovich, E. M., and A. K. Ettinger. 2014. Back to the future for plant phenology research. *New*
544 *Phytologist* 203:1021–1024.
- 545 Wright, I. J., M. Westoby, P. B. Reich, J. Oleksyn, D. D. Ackerly, Z. Baruch, F. Bongers, J. Cavender-
546 Bares, T. Chapin, J. H. C. Cornelissen, M. Diemer, J. Flexas, J. Gulias, E. Garnier, M. L. Navas,
547 C. Roumet, P. K. Groom, B. B. Lamont, K. Hikosaka, T. Lee, W. Lee, C. Lusk, J. J. Midgley,
548 Ü. Niinemets, H. Osada, H. Poorter, P. Pool, E. J. Veneklaas, L. Prior, V. I. Pyankov, S. C.
549 Thomas, M. G. Tjoelker, and R. Villar. 2004. The worldwide leaf economics spectrum. *Nature*
550 428:821–827.
- 551 Wright, J. P., and A. Sutton-Grier. 2012. Does the leaf economic spectrum hold within local species
552 pools across varying environmental conditions? *Functional Ecology* 26:1390–1398.
- 553 Yin, J., J. D. Fridley, M. S. Smith, and T. L. Bauerle. 2016. Xylem vessel traits predict the leaf
554 phenology of native and non-native understorey species of temperate deciduous forests. *Functional*
555 *Ecology* 30:206–214.
- 556 Zeng, Z. A., and E. M. Wolkovich. 2024. Weak evidence of provenance effects in spring phenology
557 across Europe and North America. *New Phytologist* 242:1957–1964.
- 558 Zohner, C. M., B. M. Benito, J.-C. Svenning, and S. S. Renner. 2016. Day length unlikely to constrain
559 climate-driven shifts in leaf-out times of northern woody plants. *Nature Climate Change* 6:1120–
560 1123.

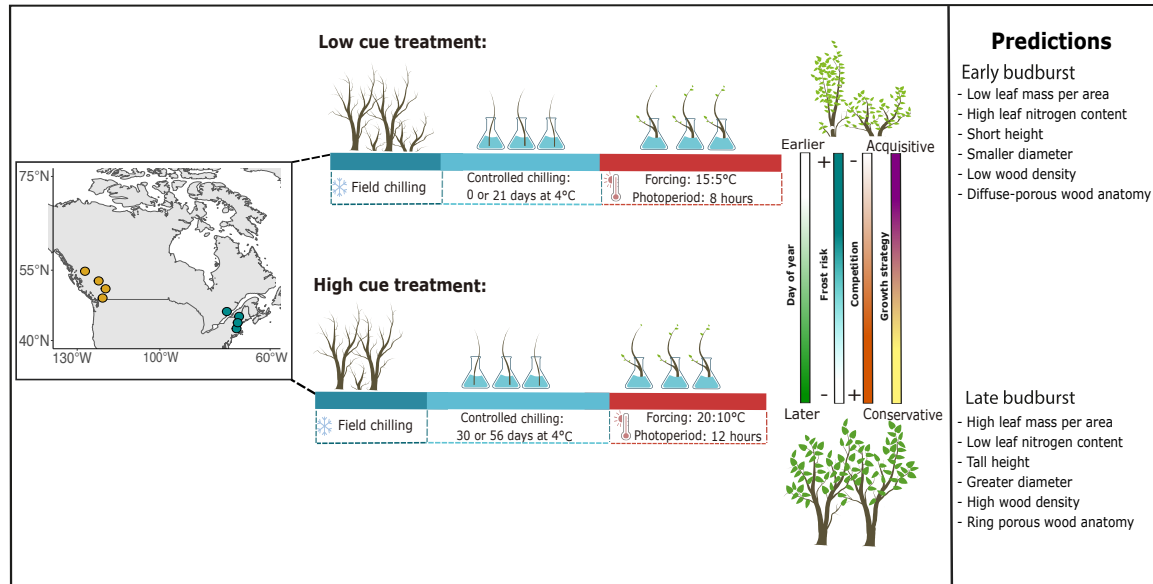


Figure 1: We collected trait data and branch cuttings from woody plants growing within eight sites, across two transects in eastern and western North America (map at left). Cuttings were used in two controlled environment (growth chamber) studies in which we applied high and low chilling, forcing, and photoperiod treatments and recorded the day of budburst. Using our paired *in situ* trait and experimental budburst data, we tested whether earlier budbursting species exhibited traits associated with more acquisitive growth strategies and smaller responses to cues and later budbursting species a more conservative growth strategy and larger responses to cues (see ‘Predictions’ at right).

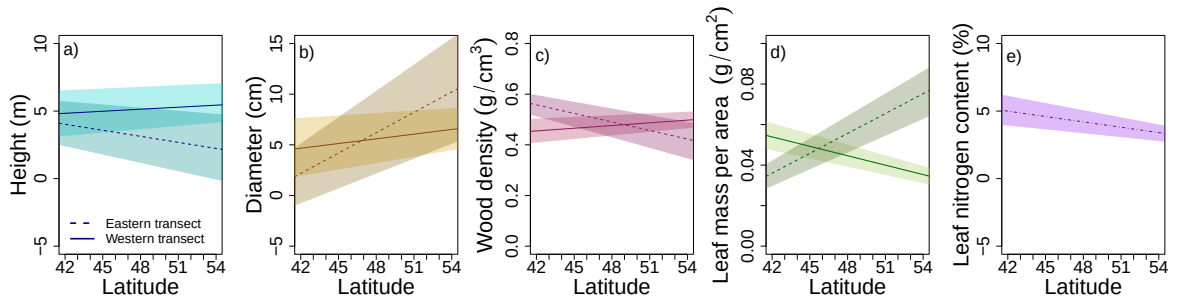


Figure 2: We found (a) height, (b) diameter, (c) branch wood density, and (d) leaf mass per area to all experience a strong interaction between latitude and transect, (e) while leaf nitrogen content showed a strong effect of latitude alone.

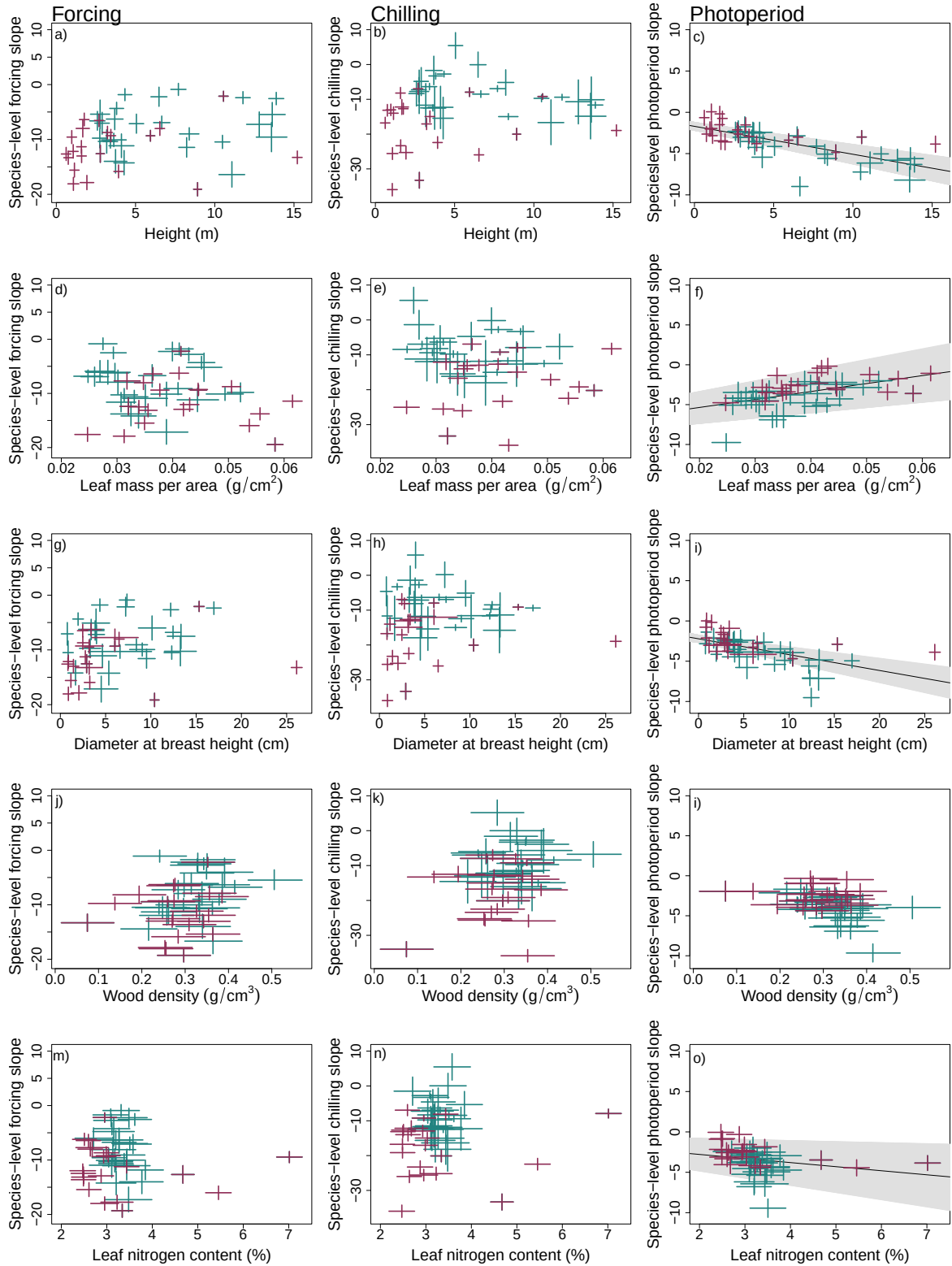


Figure 3: Relationships between species traits (rows) and cue responses (columns) showed considerable variation across: (a-c) height, (d-f) leaf mass per area, (g-i) diameter, (j-l) wood density, and (m-o) leaf nitrogen content. Point colours represent different species groups, with tree species shown in red and shrub species in blue. Crosses depict the 50% uncertainty interval of the model estimates of species trait values and estimated responses to cues. Grey bands depict large relationships between a trait and cue, representing the 90% uncertainty interval, and black lines the mean response.

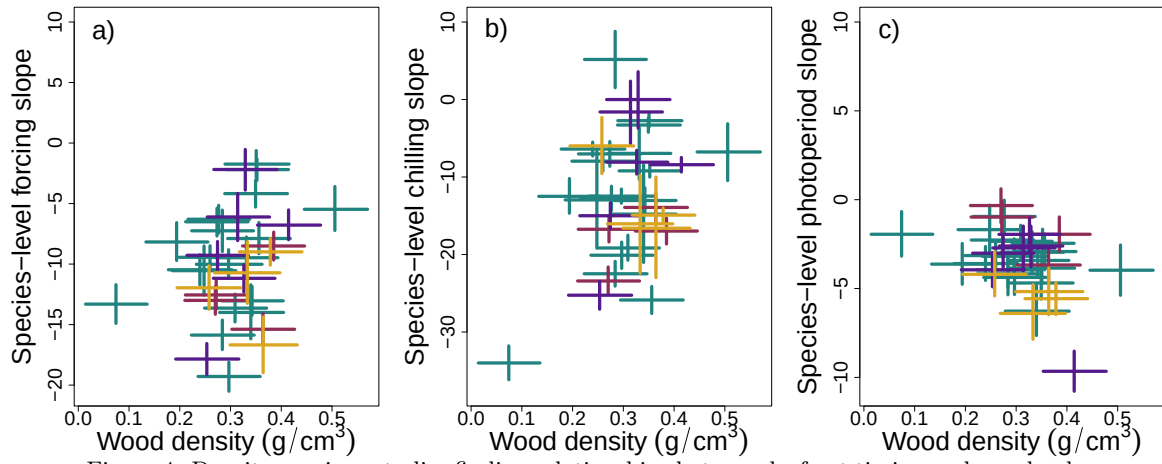


Figure 4: Despite previous studies finding relationships between leaf out timing and wood xylem structures, we did not find clear differences in species-level estimates of cue responses with wood structure or relative to their wood densities. Each cross represents the 50% uncertainty interval of (a). forcing, (b) chilling, and (c) photoperiod responses and wood density for each species, with colors depicting different types of wood structure. The lowest wood density was estimated for *Sambucus racemosa* and the highest wood density for *Viburnum lantanoides*.